Question 1

Using the mtcars dataset built in R, answer the following:

（1）Test whether the mean miles per gallon (mpg) of cars with 4 cylinders is significantly greater than 23 mpg. State your hypotheses, check the necessary assumptions, perform an appropriate one-sample test, and report your conclusion.

（2）Use permutation sampling to simulate the same one-sample test and report the empirical *p*-value. (Refer to the permutation approach for a one-sample mean discussed in the lecture materials.)

Answer：

```
# （1）Load required package and data
library(tidyverse)
data(mtcars)
# Load data
data(mtcars)

# Extract mpg for 4-cylinder cars
mpg_cyl4 <- mtcars$mpg[mtcars$cyl == 4]

# Check normality assumption
shapiro.test(mpg_cyl4)
# Shapiro-Wilk normality test
# W = 0.91244, p-value = 0.2606    -> data are consistent with normality

# Perform one-sample t-test (one-sided: greater)
t_test <- t.test(mpg_cyl4, mu = 23, alternative = "greater")
print(t_test)

# (2) Permutation simulation for one-sample mean
set.seed(123)

obs_mean <- mean(mpg_cyl4)                 # 26.66364
mu0 <- 23
centered <- mpg_cyl4 - obs_mean + mu0

n_perm <- 10000
perm_means <- replicate(n_perm, mean(sample(centered, replace = TRUE)))

# Empirical p-value: proportion of permuted means >= observed mean
emp_p <- mean(perm_means >= obs_mean)

# Histogram with full x-axis
hist(perm_means,
```

```
     xlim = range(c(perm_means, obs_mean)),
     main = "Permutation Distribution of Mean mpg (H0: mu = 23)",
     xlab = "Mean mpg", col = "lightblue", border = "white")
abline(v = obs_mean, col = "red", lwd = 2)
legend("topright", legend = c("Observed mean"), col = "red", lwd = 2)

cat("Observed mean mpg:", round(obs_mean, 3), "\n")
cat("Empirical p-value:", emp_p, "\n")
```

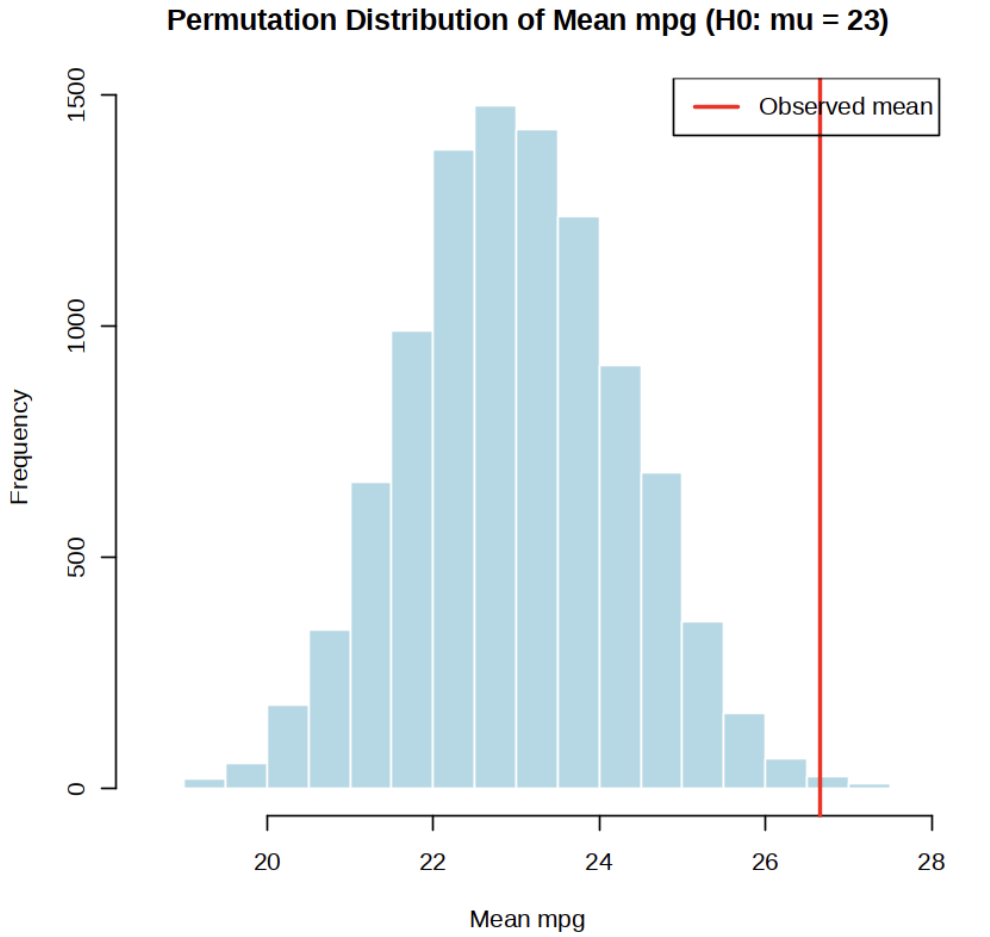
```
        Shapiro-Wilk normality test

data:  mpg_cyl4
W = 0.91244, p-value = 0.2606

        One Sample t-test

data:  mpg_cyl4
t = 2.6943, df = 10, p-value = 0.01127
alternative hypothesis: true mean is greater than 23
95 percent confidence interval:
 24.19912      Inf
sample estimates:
mean of x
 26.66364

Observed mean mpg: 26.664
Empirical p-value: 0.0028
```



Question 2

A researcher wants to determine if there is a significant improvement in student test scores after using a new study technique. The researcher recruits 15 students and measures their test scores before and after using the new study technique. Conduct a paired t-test at the 1% level of significance and plot a boxplot to visualize the distribution of the test scores before and after the treatment.

Data：

Here is the data for the test scores before and after using the new study technique:

Student	Before	After
1	75	85
2	80	82
3	70	78
4	85	88
5	65	70
6	90	95
7	78	84
8	72	80
9	88	92
10	76	81
11	84	87
12	70	73
13	68	75
14	85	90
15	77	83

Answer:

这题需要做单侧检验

样本量 n=15，显著性水平 0.01，

H0：使用新方法前后无差异，diff=0

H1：使用新方法后成绩提升，diff>0

使用 t.test 进行双样本配对检验 ：

```
##数据导入
student_data <- data.frame(
Student = 1:15,
Before = c(75, 80, 70, 85, 65, 90, 78, 72, 88, 76, 84, 70, 68, 85, 77),
After = c(85, 82, 78, 88, 70, 95, 84, 80, 92, 81, 87, 73, 75, 90, 83)
)
```

t.test(student_data$After,student_data$Before,paired = T,alternative = 'greater',conf.level =0.99)
boxplot(student_data$Before, student_data$After, names = c("Before", "After"), main = "新方法学习前后测试分数", ylab = "Score")
结果：

```
        Paired t-test

data:  student_data$After and student_data$Before
t = 9.2819, df = 14, p-value = 1.166e-07
alternative hypothesis: true mean difference is greater than 0
99 percent confidence interval:
 3.825314      Inf
sample estimates:
mean difference
       5.333333
```

p 值小于 0.01，且 t=9.281 在拒绝域内，拒绝原假设，所以在使用新的学习方法后，学生的考试成有显著的提高。

Question 3

Serum creatinine(Cr) is a biomarker for assessing kidney function, and increased levels often indicate impaired renal filtration. A pharmaceutical company is conducting a clinical trial to evaluate the renal safety of a novel anti-cancer drug. Two independent groups(Treatment vs. Placebo) of patients with comparable baseline indicators (age, sex, BMI, Cr) were enrolled. After 8 weeks, the observed data for the change in serum creatinine (ΔCr=Cr_8week-Cr_baseline, unit in mg/dL) are tested and are provided below:

ΔCr_placebo = c(0.00, 0.01, -0.03, -0.01, -0.02, -0.01, 0.03, -0.02, 0.00, 0.04, 0.02, -0.02, -0.01, 0.00, -0.03)
ΔCr_treatment = c(0.23, 0.08, 0.10, 0.15, 0.09, 0.12, 0.19, 0.08, 0.16, 0.10, 0.19, 0.12, 0.15, 0.13, 0.14)

Does the statistical evidence suggest that this drug is associated with increased renal stress?
(hint: use "var.test" and "t.test", explicitly specify the var.equal argument in your code.)

Answer:

We can use following R code to run this hypothesis test:

H0: There is no increased renal stress of this drug.
H1: There is significant increased renal stree of this drug.

First, we should test the homogeneity of variance

```
## Input data
placebo    = c(0.00, 0.01, -0.03, -0.01, -0.02, -0.01, 0.03, -0.02, 0.00, 0.04, 0.02, -0.02, -0.01, 0.00, -0.03)
treatment = c(0.23, 0.08,   0.10,   0.15,   0.09,   0.12, 0.19,   0.08, 0.16, 0.10, 0.19,   0.12,   0.15, 0.13,   0.14)
## normality test
shapiro.test(placebo) # p-value = 0.2727
shapiro.test(treatment)   # p-value = 0.426
## homogeneity of variance
var.test(placebo,treatment) # p-value = 0.008667
```

The var.test yield a p-value < 0.05, indicating that the variances are significantly different. Therefore, we should strictly set var.equal = FALSE in the t-test (Welch's t-test).

```
## Run t-test with unequal var
t.test(treatment,placebo,alternative="greater",var.equal = FALSE) # p-value = 3.037e-10
```

The t.test yield a p-value < 0.05, indicating that the drug is statistically associated with increased renal stress.

```
        Shapiro-Wilk normality test

data:  placebo
W = 0.92998, p-value = 0.2727

        Shapiro-Wilk normality test

data:  treatment
W = 0.94333, p-value = 0.426

        F test to compare two variances

data:  placebo and treatment
F = 0.22571, num df = 14, denom df = 14, p-value = 0.008667
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.07577803 0.67230140
sample estimates:
ratio of variances
         0.2257115

        Welch Two Sample t-test

data:  treatment and placebo
t = 11.011, df = 20.014, p-value = 3.037e-10
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 0.1169467       Inf
sample estimates:
   mean of x    mean of y
 0.135333333 -0.003333333
```

Question4

To study the effect of two antibiotics on the diameter (mm) of bacterial inhibition zones, independent petri dishes were randomly selected for the experiment (unpaired design). It is known that the inhibition zone diameters are normally distributed, and the variances of the two groups are equal.

Experimental group (Antibiotic A): 18.5, 19.2, 17.8, 20.1, 19.5, 18.9

Control group (Antibiotic B): 15.2, 16.8, 14.9, 16.3, 15.8, 16.1

Experimental Requirements

Test whether there is a significant difference in the antibacterial effect of the two antibiotics (two-tailed test, α=0.05);

Do not use R's built-in t.test() function;

Calculate all statistics of the two-sample unpaired t-test strictly step by step.

Answer:

group1 <- c(18.5, 19.2, 17.8, 20.1, 19.5, 18.9)

group2 <- c(15.2, 16.8, 14.9, 16.3, 15.8, 16.1)

two sides

```
# H0: μ1 = μ2
# H1: μ1 ≠ μ2
alpha <- 0.05

n1 <- length(group1)
n2 <- length(group2)

mean1 <- mean(group1)
mean2 <- mean(group2)

var1 <- var(group1)
var2 <- var(group2)

sp2 <- ((n1-1)*var1 + (n2-1)*var2) / (n1 + n2 - 2)

se <- sqrt(sp2 * (1/n1 + 1/n2))

t_stat <- (mean1 - mean2) / se

df <- n1 + n2 – 2

p_value <- 2 * pt(-abs(t_stat), df)

if (p_value < alpha) {
  cat("Conclusion: P <", alpha, ", reject the null hypothesis H0. There is a **significant difference** in the antibacterial effects of the two antibiotics.\n")
} else {
  cat("Conclusion: P ≥", alpha, ", fail to reject the null hypothesis H0. There is no significant difference in the antibacterial effects of the two antibiotics.\n")
}
```

```
Conclusion: P < 0.05 , reject the null hypothesis H0. There is a **significant difference** in the antibacterial effects of the two antibiotics.
```

Question 5

Researchers measured log10 viral loads in 10 cell lines before and after treatment with a novel interferon. Due to non-normality and a small sample size, use a non-parametric paired test to assess the treatment effect. The following table records the data for the 10 samples before and after treatment:

Sample ID	Before	After
1	6.5	5.8
2	7.2	7.0
3	5.8	5.6
4	8.1	7.2

5	6.9	6.1
6	7.5	7.6
7	6.3	5.5
8	7.8	7.1
9	7.0	6.8
10	6.7	6.2

1. Does this experiment follow an independent or paired sample design? Why is the Wilcoxon Signed-Rank Test chosen over the paired t-test in this scenario?
2. Using R, input the data above and perform the Wilcoxon Signed-Rank Test.
3. Create a Boxplot or Violin plot to visualize the changes in viral load before and after the treatment.
4. If the resulting p-value is less than 0.01, what does this represent in terms of biological research significance?

Hint: *In R, the wilcox.test() function requires the parameter paired = TRUE for paired data analysis.*

Answer:

1. Experimental Design and Method Selection

- Experimental Design: This experiment follows a paired sample design. The measurements are taken from the same 10 cell lines at two different time points (Before and After treatment).
- Method Selection: The Wilcoxon Signed-Rank Test is chosen over the paired t-test because:
    1. Non-normality: The data violates the assumption of normality required for parametric tests.
    2. Small Sample Size: With $n = 10$, parametric tests lack power and are sensitive to distribution shapes.
    3. Robustness: Wilcoxon is a non-parametric method that is more resistant to the influence of outliers.

2. R Code Implementation

You can use the following script to perform the analysis:

R

```
# Data Input
before <- c(6.5, 7.2, 5.8, 8.1, 6.9, 7.5, 6.3, 7.8, 7.0, 6.7)
after   <- c(5.8, 7.0, 5.6, 7.2, 6.1, 7.6, 5.5, 7.1, 6.8, 6.2)

# Perform Wilcoxon Signed-Rank Test
# Based on the hint: paired = TRUE is mandatory [cite: 42, 46]
test_result <- wilcox.test(before, after, paired = TRUE)
print(test_result)
```

3. Data Visualization

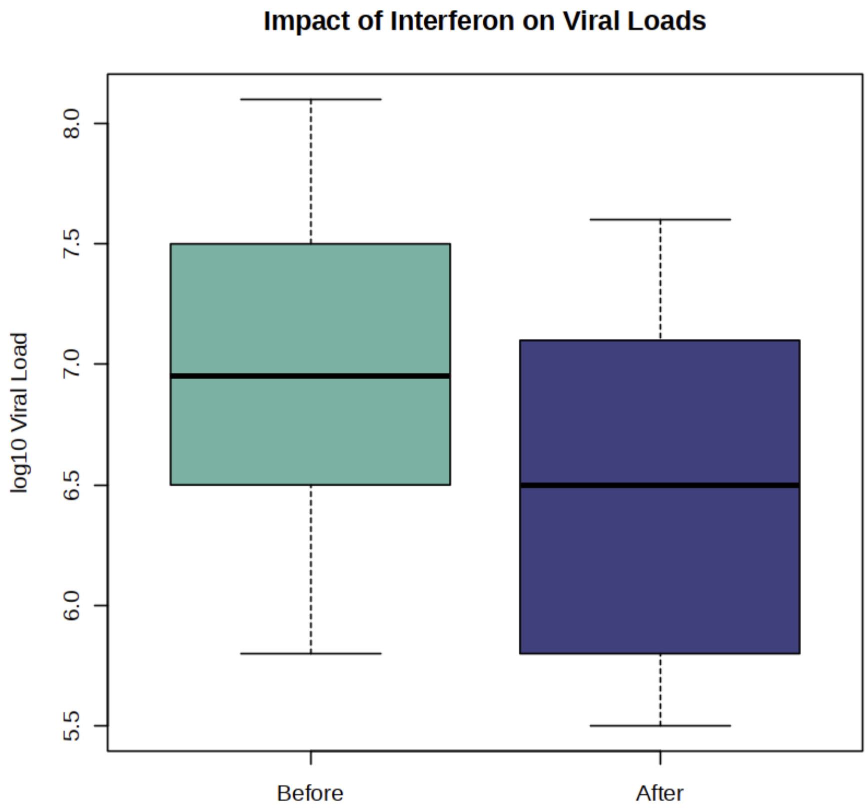
A boxplot is a standard way to visualize the distribution shift:

R

```
# Boxplot visualization
boxplot(before, after,
        names = c("Before", "After"),
        main = "Impact of Interferon on Viral Loads",
        ylab = "log10 Viral Load",
        col = c("#69b3a2", "#404080"))
```

```
	Wilcoxon signed rank test with continuity correction

data:  before and after
V = 54, p-value = 0.007842
alternative hypothesis: true location shift is not equal to 0
```



4. Biological Research Significance

Since $p = 0.007842 < 0.01$, we reject the null hypothesis. The result is highly significant. There is strong evidence that the novel interferon consistently and effectively reduces viral loads across different cell lines. This indicates a robust anti-viral effect that is highly unlikely to be due to random chance.